Control Flow Statements and Looping Statements in Python

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Aim

To write a Python program to implement control flow statements (if, if-else, if-elif-else) and looping statements (for loop, while loop, break statement, continue statement) and to study their execution and output.

Algorithm

1 Start the program.

2 Take a number and use nested if-elif-else to check whether it is positive, negative, or zero, and whether it is even or odd.

3 Use a for loop with range() to generate and print the multiplication table of a given number. 4 Use a while loop with a counter variable to compute the sum of the first N natural numbers. 5 Use a for loop with the break statement to find and print the first number in a range that is divisible by 7, then exit the loop.

6 Use a for loop with the continue statement to skip even numbers and print only the odd numbers from 1 to 10.

7 Display the output of each step.

8 Stop the program.